

Deterministic Optimization

Nonlinear Optimization
Modeling – Approximation
and Fitting

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Normal Equation and Singular
Value Decomposition

Nonlinear Optimization Modeling

Learning Objectives for this lesson

- Derive Optimality Condition for Least Squares Problem
- Discover a very important matrix factorization technique, called the Singular Value Decomposition (SVD)
- Use SVD to Solve Least Square

Optimality Condition

- Least Squares Minimization problem:

$$\min_x ||Ax - b||_2^2$$

- Equivalently: $\min_x (Ax - b)^\top (Ax - b) = x^\top A^\top Ax - 2(A^\top b)^\top x + b^\top b$
- Optimality condition:
 - $\nabla (Ax - b)^\top (Ax - b) = 0 \Rightarrow A^\top Ax = A^\top b$
- Normal equation:** $A^\top Ax = A^\top b$
- Solving least squares problem = solving Normal Equation

Solution to Normal Equation

- We wanted to “solve” $Ax = b$, but we couldn’t, because A has more rows than columns, so no such solution exists
- Instead of solving $Ax = b$, we solve $A^T Ax = A^T b$
- Since A has full column rank, $A^T A$ is invertible
- So the optimal solution to normal equation, thus to least squares problem:
 - $x = (A^T A)^{-1} A^T b$

Moore-Penrose Pseudoinverse

- For a matrix $A \in R^{m \times n}$ with $m \geq n$ and full rank, the **Moore-Penrose pseudoinverse** of A is defined as
 - $A^+ = (A^T A)^{-1} A^T$
- The solution of least squares $\min ||Ax - b||_2$ can be written as
 - $x^* = A^+ b$
 - If A is an invertible square matrix, then $A^+ = A^{-1}(A^T)^{-1}A^T = A^{-1}$
- The naïve way to compute A^+ is to first invert $A^T A$ to get $(A^T A)^{-1}$, then multiple A^T
 - However, matrix inversion can be time consuming
- There is a more efficient way: singular value decomposition

Singular Value Decomposition

- **Theorem:** Let A be an m -by- n matrix with $m \geq n$. Then we can write A as $A = U\Sigma V^\top = \sum_{i=1}^n \sigma_i u_i v_i^\top$, where
 - U is an m -by- n matrix satisfying $U^\top U = I$,
 - V is an n -by- n matrix satisfying $V^\top V = I$,
 - $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$ with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$
- Columns $u_1, \dots, u_n$ of U are called **left singular vectors** of A
- Columns $v_1, \dots, v_n$ of V are called **right singular vectors** of A
- $\sigma_1, \dots, \sigma_n$ are called **singular values** of A

Relation to Eigendecomposition

Suppose SVD of $A = U\Sigma V^\top$

- If A is a symmetric matrix with eigenvalues $\lambda_1, \dots, \lambda_n$ and orthonormal eigenvectors $u_1, \dots, u_n$, i.e. $A = U\Lambda U^\top$, where $U = [u_1, \dots, u_n]$, then $A = U\Sigma V^\top$, where $\sigma_i = |\lambda_i|$ and $v_i = \text{sign}(\lambda_i)u_i$
- Eigenvalues of $A^\top A$ are $\sigma_1^2, \dots, \sigma_n^2$, eigenvectors are $v_1, \dots, v_n$
 - $A^\top A = V\Sigma U^\top U\Sigma V^\top = V\Sigma^2 V^\top$

Use SVD to Solve Least Squares

- Solution to Least Squares: $\min ||Ax - b||_2$:
 - $x^* = A^+b = (A^T A)^{-1} A^T b$
- Since $A^T A = V \Sigma^2 V^T$, we have $(A^T A)^{-1} = V \Sigma^{-2} V^T$
- $(A^T A)^{-1} A^T = V \Sigma^{-2} V^T V \Sigma U^T = V \Sigma^{-1} U^T$
 - This can be computed efficiently after SVD
- Therefore, $x^* = V \Sigma^{-1} U^T b$

Summary

- We learned:
 - Optimality condition and normal equation for least squares problem
 - An important matrix decomposition: singular value decomposition
 - How SVD is related to eigenvalue decomposition
 - How SVD is used to solve LS